

The plus and minus of bonobo call combinations

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Berthet et al. (2025) propose a method to understand animal communication that constitutes a pioneering endeavour. Starting from a joint ethogram of events and vocal signals of bonobos, they propose criteria to infer, bottom-up, how the meanings of these calls are combined in sequences—an extraordinarily ambitious program. The proposed criteria (partly coming from Trujillo and Holler 2024), however, are still imperfect.

The criteria generate false negatives: because they assume that composite meanings are close to their parts (criterion (iii)), they miss paradigmatic cases of non-trivial compositionality: the meaning of “bad” may not be close to “bad dancer”, not closer than “singer” or “music” is. Negation, the flagship of non-trivial compositionality, provides another example: “dancer” alone is by construction incompatible with “non-dancer”.

The criteria also generate false positives: we tested all possible trivial compositions of meanings (<https://github.com/emmanuelchemla/callsfromfeatures/>). None were recognized as such by the current criteria; 23% were classified as non-trivial, at the risk of overestimating the complexity of the system. This results from the attractive but unreliable use of addition in semantic space (Linzen 2016), here as a proxy for conjunction.

Simulations can evaluate several possible production rules, including “Send all appropriate signals” (Schlenker et al, in press). Under this rule, simple conjunctions unfortunately appear non-trivial (70% of the cases). That is because calls A, B and A&B will never overlap, and conjunction is not anymore the intersection operation that it should be. Thus, tests must jointly account for various assumed properties of communication.

The behavioral data result from a generation process involving many interacting properties: natural frequencies of events, meanings, compositional and production rules. By modelling them explicitly, we can reverse-engineer *all of* these properties and follow the path opened by Berthet et al. (2025) to understand animal communication through scalable, data-driven methods.

M. Berthet et al. Extensive compositionality in the vocal system of bonobos. *Science* **388**, 104-108 (2025). DOI:[10.1126/science.adv1170](https://doi.org/10.1126/science.adv1170)

Tal Linzen. 2016. [Issues in evaluating semantic spaces using word analogies](#). In *Proceedings of the 1st Workshop on Evaluating Vector-Space Representations for NLP*, pages 13–18, Berlin, Germany. Association for Computational Linguistics.

P. Schlenker et al., Ancestral Meanings: a Prelude to Evolutionary Animal Linguistics. *Linguistics & Philosophy*, in press. <https://ling.auf.net/lingbuzz/008203>

J. P. Trujillo, J. Holler, Conversational facial signals combine into compositional meanings that change the interpretation of speaker intentions. *Sci. Rep.* **14**, 2286 (2024).